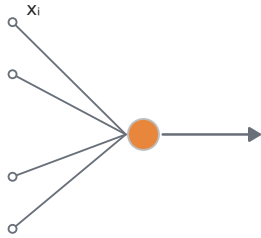
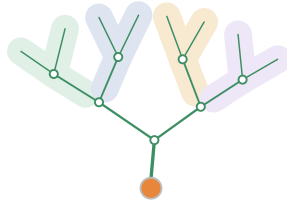
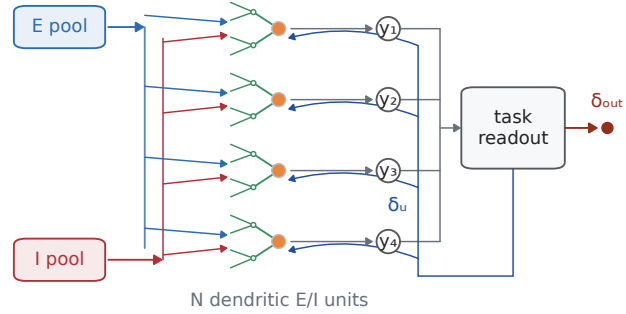
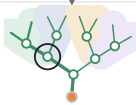
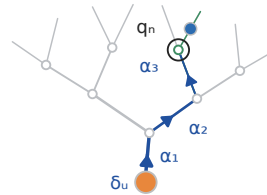


APoint unit $\rightarrow$ dendritic treeone neuronal coordinate δ_u subtree coefficients $C_{u,k}$ **B**

Network layer

**C**Coordinate $\rightarrow$ address $\rightarrow$ gaincoordinate
which neuron δ_u address
which subtree $C_{u,k}$ gain
how strongly $\tilde{\alpha}_n$ **D**Local eligibility $\times$ transported errordirected-tree eligibility e_i • x_i presyn. $\curvearrowright E_i - V_n$ $\sim R_n^{\text{tot}}$ directed tree: $\partial \mathcal{L} / \partial g_i = [x_i R_n^{\text{tot}} (E_i - V_n)] [\partial \mathcal{L} / \partial V_n]$ general adjoint: $\partial \mathcal{L} / \partial g_i = x_i (E_i - V_n) q_n$, $q_n = R_n^{\text{tot}} \partial \mathcal{L} / \partial V_n$ **E**

Roadmap of the Results

